

Unit 13: The Chain Rule — Full Solutions

Warm-up

1. $y = (x + 3)^4$

Outer layer: something to the 4th power. Inner layer: $u = x + 3$, so $u' = 1$.

$$y' = 4(x + 3)^3 \cdot 1 = 4(x + 3)^3$$

2. $y = (2x - 1)^5$

Inner: $u = 2x - 1$, $u' = 2$.

$$y' = 5(2x - 1)^4 \cdot 2 = 10(2x - 1)^4$$

3. $y = (3x + 2)^3$

Inner: $u = 3x + 2$, $u' = 3$.

$$y' = 3(3x + 2)^2 \cdot 3 = 9(3x + 2)^2$$

4. $y = (x^2 + 1)^3$

Inner: $u = x^2 + 1$, $u' = 2x$.

$$y' = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$$

5. $y = (5 - x)^4$

Inner: $u = 5 - x$, $u' = -1$.

$$y' = 4(5 - x)^3 \cdot (-1) = -4(5 - x)^3$$

6. $y = (4x + 7)^2$

Inner: $u = 4x + 7$, $u' = 4$.

$$y' = 2(4x + 7) \cdot 4 = 8(4x + 7)$$

7. $y = (x^2 - 2x)^2$

Inner: $u = x^2 - 2x$, $u' = 2x - 2$.

$$y' = 2(x^2 - 2x) \cdot (2x - 2)$$

We can factor this further: $x^2 - 2x = x(x - 2)$ and $2x - 2 = 2(x - 1)$, so:

$$y' = 2 \cdot x(x-2) \cdot 2(x-1) = 4x(x-1)(x-2)$$

$$y' = 4x(x-1)(x-2).$$

Standard

8. $y = \sin(3x)$

Inner: $u = 3x, u' = 3.$

$$y' = \cos(3x) \cdot 3 = 3\cos(3x)$$

9. $y = \cos(x^2)$

Inner: $u = x^2, u' = 2x.$

$$y' = -\sin(x^2) \cdot 2x = -2x\sin(x^2)$$

10. $y = (2x^2 - 3)^{-1}$

Inner: $u = 2x^2 - 3, u' = 4x.$

$$y' = -1 \cdot (2x^2 - 3)^{-2} \cdot 4x = \frac{-4x}{(2x^2 - 3)^2}$$

11. $y = \sqrt{x^2 + 4} = (x^2 + 4)^{1/2}$

Inner: $u = x^2 + 4, u' = 2x.$

$$y' = \frac{1}{2}(x^2 + 4)^{-1/2} \cdot 2x = x(x^2 + 4)^{-1/2} = \frac{x}{\sqrt{x^2 + 4}}$$

12. $y = \tan(2x)$

Inner: $u = 2x, u' = 2.$

$$y' = \sec^2(2x) \cdot 2 = 2\sec^2(2x)$$

13. $y = (3x - 1)^{1/3}$

Inner: $u = 3x - 1, u' = 3.$

$$y' = \frac{1}{3}(3x - 1)^{-2/3} \cdot 3 = (3x - 1)^{-2/3} = \frac{1}{(3x - 1)^{2/3}}$$

14. $y = \sin^2 x = (\sin x)^2$

Inner: $u = \sin x$, $u' = \cos x$.

$$y' = 2(\sin x)^1 \cdot \cos x = 2 \sin x \cos x$$

Challenge

15. $y = \left(\frac{x^2}{2} + x - \frac{1}{x}\right)^4$

First, find the derivative of the inner expression. Rewrite $-\frac{1}{x}$ as $-x^{-1}$:

$$u = \frac{x^2}{2} + x - x^{-1}$$

$$u' = x + 1 - (-1)x^{-2} = x + 1 + x^{-2} = x + 1 + \frac{1}{x^2}$$

Now apply the power chain rule (outer power is 4):

$$y' = 4 \left(\frac{x^2}{2} + x - \frac{1}{x}\right)^3 \cdot \left(x + 1 + \frac{1}{x^2}\right)$$

$$y' = 4 \left(\frac{x^2}{2} + x - \frac{1}{x}\right)^3 \left(x + 1 + \frac{1}{x^2}\right).$$

16. $y = \sin^4 x = (\sin x)^4$

Inner: $u = \sin x$, $u' = \cos x$.

$$y' = 4(\sin x)^3 \cdot \cos x = 4 \sin^3 x \cos x$$

17. $y = x^2 \sin^4 x + x \cos^{-2} x$

This has two terms, and each one needs the product rule (since each is a plain factor times a trig-composition factor).

Term 1: $x^2 \sin^4 x$. Let $f = x^2$ ($f' = 2x$) and $g = \sin^4 x$ (from Problem 16, $g' = 4 \sin^3 x \cos x$).

$$\frac{d}{dx}[x^2 \sin^4 x] = 2x \sin^4 x + x^2 \cdot 4 \sin^3 x \cos x = 2x \sin^4 x + 4x^2 \sin^3 x \cos x$$

Term 2: $x \cos^{-2} x$. Let $f = x$ ($f' = 1$) and $g = \cos^{-2} x = (\cos x)^{-2}$.

Find g' using the power chain rule, with inner $u = \cos x$ ($u' = -\sin x$):

$$g' = -2(\cos x)^{-3} \cdot (-\sin x) = 2 \sin x (\cos x)^{-3} = \frac{2 \sin x}{\cos^3 x}$$

$$\frac{d}{dx}[x \cos^{-2} x] = 1 \cdot \cos^{-2} x + x \cdot \frac{2 \sin x}{\cos^3 x} = \sec^2 x + \frac{2x \sin x}{\cos^3 x}$$

Note $\frac{2x \sin x}{\cos^3 x} = 2x \left(\frac{\sin x}{\cos x} \right) \left(\frac{1}{\cos^2 x} \right) = 2x \tan x \sec^2 x$.

Adding both terms together:

$$y' = (2x \sin^4 x + 4x^2 \sin^3 x \cos x) + (\sec^2 x + 2x \tan x \sec^2 x)$$

$$y' = 2x \sin^4 x + 4x^2 \sin^3 x \cos x + \sec^2 x + 2x \tan x \sec^2 x.$$

18. $y = (x^2 + 1)^3(x - 1)^2$

Let $f = (x^2 + 1)^3$. Using the power chain rule (inner $u = x^2 + 1$, $u' = 2x$):

$$f' = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$$

Let $g = (x - 1)^2$, so $g' = 2(x - 1)$.

Apply the product rule:

$$y' = f'g + fg' = 6x(x^2 + 1)^2(x - 1)^2 + (x^2 + 1)^3 \cdot 2(x - 1)$$

Both terms share a common factor of $(x^2 + 1)^2(x - 1)$. Factor it out:

$$y' = (x^2 + 1)^2(x - 1) \left[6x(x - 1) + 2(x^2 + 1) \right]$$

Simplify inside the brackets: $6x(x - 1) = 6x^2 - 6x$, and adding $2(x^2 + 1) = 2x^2 + 2$ gives $6x^2 - 6x + 2x^2 + 2 = 8x^2 - 6x + 2$.

$$y' = (x^2 + 1)^2(x - 1)(8x^2 - 6x + 2)$$

We can factor out a 2 from the last piece:

$$y' = 2(x^2 + 1)^2(x - 1)(4x^2 - 3x + 1)$$

$$y' = 2(x^2 + 1)^2(x - 1)(4x^2 - 3x + 1).$$

19. $y = \sqrt{\sin x} = (\sin x)^{1/2}$

Inner: $u = \sin x$, $u' = \cos x$.

$$y' = \frac{1}{2}(\sin x)^{-1/2} \cdot \cos x = \frac{\cos x}{2\sqrt{\sin x}}$$

20. $y = (1 + \cos x)^3$

Inner: $u = 1 + \cos x$, $u' = -\sin x$.

$$y' = 3(1 + \cos x)^2 \cdot (-\sin x) = -3 \sin x (1 + \cos x)^2$$

Applied

21. $V(r) = \frac{4}{3}\pi r^3$, where $r(t) = 2t$.

Using the chain rule, $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$.

$$\frac{dV}{dr} = 4\pi r^2 \quad \frac{dr}{dt} = 2$$

$$\frac{dV}{dt} = 4\pi r^2 \cdot 2 = 8\pi r^2$$

At $t = 1$: $r(1) = 2(1) = 2$, so:

$$\left. \frac{dV}{dt} \right|_{t=1} = 8\pi(2)^2 = 8\pi(4) = 32\pi$$

Answer: $\frac{dV}{dt} = 8\pi r^2$, and at $t = 1$, $\frac{dV}{dt} = 32\pi$ cubic cm per unit time.

22. $h(t) = 3\sin(2t)$

Inner: $u = 2t$, $u' = 2$.

$$h'(t) = 3\cos(2t) \cdot 2 = 6\cos(2t)$$

$$h'(0) = 6\cos(0) = 6(1) = 6$$

Answer: $h'(t) = 6\cos(2t)$, and $h'(0) = 6$. At $t = 0$, the wave height is increasing at a rate of 6 cm per unit time.

23. $C(x) = \sqrt{x^2 + 50}$, where $x(t) = 10t$.

Using the chain rule, $\frac{dC}{dt} = \frac{dC}{dx} \cdot \frac{dx}{dt}$.

$$\frac{dC}{dx} = \frac{1}{2}(x^2 + 50)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + 50}} \quad \frac{dx}{dt} = 10$$

$$\frac{dC}{dt} = \frac{x}{\sqrt{x^2 + 50}} \cdot 10$$

At $t = 2$: $x(2) = 10(2) = 20$.

$$\sqrt{20^2 + 50} = \sqrt{400 + 50} = \sqrt{450} = \sqrt{225 \cdot 2} = 15\sqrt{2}$$

$$\left. \frac{dC}{dt} \right|_{t=2} = \frac{20}{15\sqrt{2}} \cdot 10 = \frac{200}{15\sqrt{2}} = \frac{40}{3\sqrt{2}} = \frac{40\sqrt{2}}{6} = \frac{20\sqrt{2}}{3}$$

Answer: at $t = 2$, $\frac{dC}{dt} = \frac{20\sqrt{2}}{3} \approx 9.43$ dollars per unit time.

24. $T(t) = (t+2)^3 - 5$

Inner: $u = t + 2$, $u' = 1$.

$$T'(t) = 3(t+2)^2 \cdot 1 = 3(t+2)^2$$

$$T'(1) = 3(1+2)^2 = 3(9) = 27$$

Answer: $T'(t) = 3(t+2)^2$, and $T'(1) = 27$. At $t = 1$, the greenhouse temperature is increasing at a rate of $27^\circ F$ per unit time.